# A short note on Dupire's formula

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**Abstract**

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## The Dupire Equation

For each strike $K$ and maturity $T$, we need to solve backward parabolic Black-Scholes PDE in the variables $(S, t)$, to find the option prices. It is possible to find the same option price by solving a dual problem, namely a forward equation in the variables $(K, T)$, given the present spot value and time known as the *dual Black-Scholes equation* or Dupire's PDE.

It is not straight-forward to understand how to extract a surface $\sigma_{LV}$ from the market. Dupire succeeded to prove that assuming a continuum of traded plain vanilla options on the market, there exists one and only one local volatility surface $\sigma_{LV}$ (a unique diffusion process) such that model is perfectly able to reproduce the current market prices.

## Derivation of the Dupire's Formula

**Proposition 1.** *(Dupire's Equation) The value of a call option as a function of the strike price $K$ and the time to maturity $T$ given the present value of the stock $S$ is given by the following forward parabolic equation known as the Dupire's equation:*

Suppose that the stock price diffuses with the risk-neutral drift $\mu_t = r - q$ and local volatility $\sigma(S, t)$ according to the dynamics:

$$dS_t = \mu S_t dt + \sigma(S_t, t) S_t dB_t \tag{1}$$

Suppose the stock price today at $t = 0$ is $S_0$. Consider the collection of undiscounted option prices of options struck at $K$ and expiration $T$, $\{C(S_0, K, T)\}$. It is given by:

$$C(S_0, K, T) = \int_{\mathbb{R}} (S_T - K)^+ p(S_T, T | S_0, 0) dS_T \tag{2}$$

which is an expectation over the state space of $S_T$ and $p(S_T, T|S_0, 0)$ is the transition probability density from state $(S_0, 0)$ to state $(S_T, T)$.

The probability density $p(S_T, T|S_0, 0)$ satisfies the Fokker-Planck equation (forward PDE):

$$\frac{\partial p_T}{\partial T} = A^* p_T \tag{3}$$

where $A^*$ is the adjoint of the generator of the diffusion $(S_t, t \geq 0)$. In other words:

$$\frac{\partial p}{\partial T} = \frac{1}{2} \frac{\partial^2}{\partial S_T^2} (\sigma(S_T)^2 S_T^2 p) - \frac{\partial}{\partial S_T} ((r - q) S_T p) \tag{4}$$

Differentiating (2) with respect to $K$ gives:

$$
\begin{aligned}
\frac{\partial C}{\partial K} &= \frac{\partial}{\partial K}\left(\int_{\mathbb{R}}(S_T-K)^+p(S_T,T|S_0)dS_T\right)\\
&= \frac{\partial}{\partial K}\left(\int_K^{\infty}(S_T-K)p(S_T,T|S_0)dS_T\right)\\
&= \frac{\partial}{\partial K}\left(\int_{\alpha(K)}^{\beta(K)}f(S_T,K)dS_T\right)\\
&= \underbrace{f(\beta(K),K)\beta'(K)}_{\lim_{\beta(K)\to\infty}p(\beta(K),T)=0} - \underbrace{f(\alpha(K),K)\alpha'(K)}_{f(\alpha(K),K)=(K-K)p(K,T)=0} + \int_{\alpha(K)}^{\beta(K)}\frac{\partial}{\partial K}(f(S_T,K))dS_T\\
&= \int_K^{\infty}\frac{\partial}{\partial K}(S_T-K)p(S_T,T|S_0)dS_T\\
&= -\int_K^{\infty}p(S_T,T|S_0)dS_T \qquad (5)
\end{aligned}
$$

Differentiating again with respect to $K$, we have:

$$
\begin{aligned}
\frac{\partial^2 C}{\partial K^2} &= -\frac{\partial}{\partial K}\int_K^{\infty}p(S_T,T|S_0)dS_T\\
&= -\left[\underbrace{p(\beta(K),T)\beta'(K)}_{0} - p(\alpha(K),T)\alpha'(K) + \int_K^{\infty}\underbrace{\frac{\partial}{\partial K}p(S_T,T)}_{0}dS_T\right]\\
&= p(K,T) \qquad (6)
\end{aligned}
$$

Now, differentiating the undiscounted option price in (2) with respect to $T$, yields:

$$
\begin{aligned}
\frac{\partial C}{\partial T} &= \frac{\partial}{\partial T}\left(\int_K^{\infty}(S_T-K)p(S_T,T|S_0,0)dS_T\right)\\
&= \int_K^{\infty}(S_T-K)\left\{\frac{\partial}{\partial T}(p(S_T,T|S_0,0))\right\}dS_T\\
&= \int_K^{\infty}(S_T-K)\left\{\frac{1}{2}\frac{\partial^2}{\partial S_T^2}(\sigma(S_T)^2S_T^2p_T(S_T,T|S_0,0)) - \frac{\partial}{\partial S_T}((r-q)S_Tp_T(S_T,T|S_0,0))\right\}dS_T \qquad (7)\\
&= -(r-q)I_1 + \frac{1}{2}I_2 \qquad (8)
\end{aligned}
$$

where the short notation $I_1$ and $I_2$ is introduced for the integrals in the last-but-one line.

## First identity

From the definition of the Call option price, we have:

$$
C = \int_K^{\infty}(S_T-K)p_T(S_T,T|S_0,0)dS_T
$$

On the other hand from equation (5), we have:

$$
\frac{\partial C}{\partial K} = -\int_K^{\infty}p(S_T,T|S_0,0)dS_T
$$

So,

$$
\begin{aligned}
C - K\frac{\partial C}{\partial K} &= \int_K^\infty (S_T - K)p_T(S_T, T|S_0, 0) + Kp_T(S_T, T|S_0, 0)dS_T \\
&= \int_K^\infty S_T p(S_T, T|S_0, 0)dS_T
\end{aligned} \tag{9}
$$

## Evaluating the integrals $I_1$ and $I_2$

We can now proceed with evaluating the integrals $I_1$ and $I_2$. For $I_1$, using integration by parts, we get:

$$
\begin{aligned}
I_1 &= \int_K^\infty (S_T - K)\frac{\partial}{\partial S_T}(S_T p(S_T, T|S_0, 0))dS_T \\
&= [(S_T - K)S_T p(S_T, T|S_0, 0)]_K^\infty - \int_K^\infty S_T p(S_T, T|S_0, 0)\frac{\partial}{\partial S_T}(S_T - K) \cdot dS_T \\
&= -\int_K^\infty S_T p(S_T, T|S_0, 0)dS_T \\
&= K\frac{\partial C}{\partial K} - C
\end{aligned} \tag{10}
$$

where in the last line, we used the result from (9).

For $I_2$, using integration by parts, we obtain:

$$
\begin{aligned}
I_2 &= \int_K^\infty (S_T - K)\frac{\partial^2}{\partial S_T^2}[\sigma^2 S_T^2 p(S_T, T|S_0, 0)]dS_T \\
&= \left[(S_T - K)\frac{\partial}{\partial S_T}[\sigma(S_T, T)^2 S_T^2 p(S_T, T|S_0, 0)]\right]_K^\infty - \int_K^\infty \frac{\partial}{\partial S_T}[\sigma(S_T, T)^2 S_T^2 p(S_T, T|S_0, 0)]dS_T
\end{aligned}
$$

It is assumed that first term evaluates to 0; whereas for the second term, observe that differentiation and integration are inverse operations. So:

$$
\begin{aligned}
I_2 &= -\left[\sigma(S_T, T)^2 S_T^2 p(S_T, T|S_0, 0)\right]_K^\infty \\
&= \sigma(K, T)^2 K^2 p(K, T|S_0, 0) \\
&= \sigma(K, T)^2 K^2 \frac{\partial^2 C}{\partial K^2}
\end{aligned} \tag{11}
$$

## The final step

Substituting $I_1$ and $I_2$ into (8), we have the Dupire PDE:

$$
\frac{\partial C}{\partial T} = -(r - q)\left(K\frac{\partial C}{\partial K} - C\right) + \frac{1}{2}\sigma(K, T)^2\frac{\partial^2 C}{\partial K^2} \tag{12}
$$

The equation can also be solved for the local volatility $\sigma(S_t, t)$ at the point $(K, T)$:

$$
\sigma^2(K, T) = \frac{\frac{\partial C}{\partial T} + (r - q)\left(K\frac{\partial C}{\partial K} - C\right)}{\frac{1}{2}K^2\frac{\partial^2 C}{\partial K^2}}
$$

The right-hand side of this equation can be computed from known European option prices. So, given a complete set of European option prices for all strikes and expirations, local volatilities are uniquely given by the above equation.